Certificate of Analysis

Olink® Explore

PROJECT NAME	20201991_Appelmans_Plasma_NEURO
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CONTACT	Eline Appelmans
	eappelmans@fnih.org
BUSINESS DEVELOPMENT MANAGER	Josh Porter
	josh.porter@olink.com
ANALYSIS LAB	Olink
	service@olink.com

1. Project information

No. of samples	No. of plates	Normalization method
746	9	Intensity normalization

1.1 Sample type

EDTA_PLASMA

1.2 Project specific comments

N/A

2. Quality control

Three internal controls are added to each sample and two of them are used to monitor the quality of assay performance, as well as the quality of individual samples:

1. Incubation control
2. Amplification Control

The following parameters are evaluated in the Quality Control (QC):

1. The average matched counts[1] for each sample. To pass QC, there should be at least 500 counts, otherwise the sample receives a QC warning status.
2. The deviation from the median value of the incubation- and amplification controls for each individual sample. To pass QC, the deviation should not exceed +/-0.3 NPX for either of the internal controls, otherwise the sample will receive a QC warning status.

[1] The number of reads for each specific combination of sample and assay

Data from all samples is included in the data output file. Samples that did not pass the QC are indicated in columns named "QC warning". Data points from samples that do not pass QC should be treated with caution.

2.1 QC summary

Olink Panel	No. of samples that passed QC / Total no. of samples	Passed samples (%)
Explore 384 Neurology	597 / 746	80

2.2 Intra- and Inter-assay Coefficient of Variance (%CV)

Intra- and inter-CVs are based on the control samples (pooled plasma samples) included on each sample plate. Calculations are made for each assay using NPX-values. Average % CV for all assays on a panel is presented in section 2.2.1. The number of assays with CVs within defined intervals are presented in sections 2.2.2 and 2.2.3.

2.2.1 Average %CV

Olink Panel	Intra-assay %CV	Inter-assay %CV
Explore 384 Neurology	16	21

2.2.2 Intra-assay %CV distribution

Olink Panel	<5%	5-10%	10-15%	>15%	N/A*
Explore 384 Neurology	22	65	88	144	48

*Assays where CV is not possible to calculate

2.2.3 Inter-assay %CV distribution

Olink Panel	<10%	10-20%	20-30%	>30%	N/A*
Explore 384 Neurology	22	130	158	23	34

*Assays where CV is not possible to calculate

3. Protein detection results

3.1 Number of proteins detected in >50% of the samples

Olink Panel	No. of detected proteins / Total no. of proteins	Detected proteins (%)	Expected detectability in EDTA plasma* (%)
Explore 384 Neurology	334 / 367	91	N/A

*The expected detectability is based on EDTA plasma from healthy donors. These values are intended as guidelines only and protein levels are expected to vary depending on different pathological conditions, sample matrices, or sample preparation methods.

3.2 Data output

Data is presented as NPX (Normalized Protein eXpression) values. NPX is Olink's relative protein quantification unit on log2 scale. NPX values from Olink® Explore 384 and Olink® Explore 1536 (a combination of 4 separate Olink® Explore 384 panels) are calculated from the number of matched counts, using NGS (Next Generation Sequencing) as readout. The NPX values are presented in a separate results file delivered in the MyData cloud. Data values for measurements below limit of detection (LOD) are reported for all samples.